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Assignment 6

Exercise 8.3-1 Using Figure 8.3 as a model, illustrate the operation of RADIX-SORT on the following list of English words: COW, DOG, SEA, RUG, ROW, MOB, BOX, TAB, BAR, EAR, TAR, DIG, BIG, TEA, NOW, FOX.

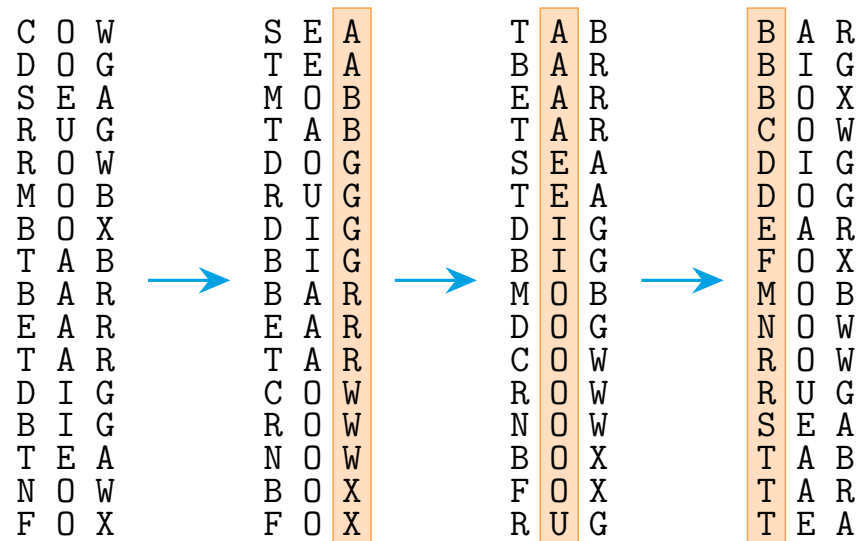
Solution

Figure 1: Visualization of RADIX-SORT for provided input

Exercise 14.2-1 Find an optimal parenthesization of a matrix-chain product whose sequence of dimensions is $\langle 5, 10, 3, 12, 5, 50, 6 \rangle$.

Solution

Based on the following information, the matrices looks as following:

A_1	A_2	A_3	A_4	A_5	A_6
5×10	10×3	3×12	12×5	5×50	50×6

Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[i, j] = \begin{cases} 0 & \text{if } i = j \\ \min_{i \leq k < j} \{m[i, k] + m[k + 1, j] + p_{i-1}p_kp_j\} & \text{if } i < j \end{cases}$$

We know from the start, that the matrices alone result in 0

$$m[1, 1] = m[2, 2] = m[3, 3] = m[4, 4] = m[5, 5] = m[6, 6] = 0$$

So we can prefill the table:

m		j					
		1	2	3	4	5	6
i	1	0					
	2	—	0				
	3	—	—	0			
	4	—	—	—	0		
	5	—	—	—	—	0	
	6	—	—	—	—	—	0

Then we calculate with two adjacent matrices ($\Delta = 1$):

$$m[1, 2] = 0 + 0 + (5 \times 10 \times 3) = 150$$

$$m[2, 3] = 0 + 0 + (10 \times 3 \times 12) = 360$$

$$m[3, 4] = 0 + 0 + (3 \times 12 \times 5) = 180$$

$$m[4, 5] = 0 + 0 + (12 \times 5 \times 50) = 3000$$

$$m[5, 6] = 0 + 0 + (5 \times 50 \times 6) = 1500$$

After which, we can update the table:

m		j					
		1	2	3	4	5	6
i	1	0	150				
	2	—	0	360			
	3	—	—	0	180		
	4	—	—	—	0	3000	
	5	—	—	—	—	0	1500
	6	—	—	—	—	—	0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[1, 3] = \begin{cases} m[1, 1] + m[2, 3] + 5 \times 10 \times 12 = 960 \\ m[1, 2] + m[3, 3] + 5 \times 3 \times 12 = 330 \end{cases}$$

After which, we can update the table:

m	j						
	1	2	3	4	5	6	
i	1	0	150	330			
	2	—	0	360			
	3	—	—	0	180		
	4	—	—	—	0	3000	
	5	—	—	—	—	0	1500
	6	—	—	—	—	—	0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[2, 4] = \begin{cases} \cancel{m[2, 2]}^0 + \cancel{m[3, 4]}^{180} + 10 \times 3 \times 5 & = 330 \\ \cancel{m[2, 3]}^{360} + \cancel{m[4, 4]}^0 + 10 \times 12 \times 5 & = 960 \end{cases}$$

After which, we can update the table:

m	j						
	1	2	3	4	5	6	
i	1	0	150	330			
	2	—	0	360	330		
	3	—	—	0	180		
	4	—	—	—	0	3000	
	5	—	—	—	—	0	1500
	6	—	—	—	—	—	0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[3, 5] = \begin{cases} \cancel{m[3, 3]}^0 + \cancel{m[4, 5]}^{3000} + 3 \times 12 \times 50 & = 4800 \\ \cancel{m[3, 4]}^{180} + \cancel{m[5, 5]}^0 + 3 \times 5 \times 50 & = 930 \end{cases}$$

After which, we can update the table:

m	j						
	1	2	3	4	5	6	
i	1	0	150	330			
	2	—	0	360	330		
	3	—	—	0	180	930	
	4	—	—	—	0	3000	
	5	—	—	—	—	0	1500
	6	—	—	—	—	—	0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following

formula:

$$m[4, 6] = \begin{cases} \cancel{m[4, 4]}^0 + \cancel{m[5, 6]}^{1500} + 12 \times 5 \times 6 & = 1860 \\ \cancel{m[4, 5]}^{3000} + \cancel{m[6, 6]}^0 + 12 \times 50 \times 6 & = 6600 \end{cases}$$

After which, we can update the table:

m	j					
	1	2	3	4	5	6
i	1	0	150	330		
	2	—	0	360	330	
	3	—	—	0	180	930
	4	—	—	—	0	3000 1860
	5	—	—	—	—	0 1500
	6	—	—	—	—	— 0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[1, 4] = \begin{cases} \cancel{m[1, 1]}^0 + \cancel{m[2, 4]}^{330} + 5 \times 10 \times 5 & = 580 \\ \cancel{m[1, 2]}^{150} + \cancel{m[3, 4]}^{180} + 5 \times 3 \times 5 & = 405 \\ \cancel{m[1, 3]}^{330} + \cancel{m[4, 4]}^0 + 5 \times 12 \times 5 & = 630 \end{cases}$$

After which, we can update the table:

m	j					
	1	2	3	4	5	6
i	1	0	150	330	630	
	2	—	0	360	330	
	3	—	—	0	180	930
	4	—	—	—	0	3000 1860
	5	—	—	—	—	0 1500
	6	—	—	—	—	— 0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[2, 5] = \begin{cases} \cancel{m[2, 2]}^0 + \cancel{m[3, 5]}^{930} + 10 \times 3 \times 50 & = 2430 \\ \cancel{m[2, 3]}^{360} + \cancel{m[4, 5]}^{3000} + 10 \times 12 \times 50 & = 9360 \\ \cancel{m[2, 4]}^{330} + \cancel{m[5, 5]}^0 + 10 \times 5 \times 50 & = 2830 \end{cases}$$

After which, we can update the table:

m	j					
	1	2	3	4	5	6
i	1	0	150	330	630	
	2	—	0	360	330	2430
	3	—	—	0	180	930
	4	—	—	—	0	3000 1860
	5	—	—	—	—	0 1500
	6	—	—	—	—	— 0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[3, 6] = \begin{cases} \cancel{m[3, 3]}^0 + \cancel{m[4, 6]}^{1860} + 3 \times 12 \times 6 = 2076 \\ \cancel{m[3, 4]}^{180} + \cancel{m[5, 6]}^{1500} + 3 \times 5 \times 6 = 1770 \\ \cancel{m[3, 5]}^{930} + \cancel{m[6, 6]}^0 + 3 \times 50 \times 6 = 1830 \end{cases}$$

After which, we can update the table:

m	j					
	1	2	3	4	5	6
i	1	0	150	330	630	
	2	—	0	360	330	2430
	3	—	—	0	180	930 1770
	4	—	—	—	0	3000 1860
	5	—	—	—	—	0 1500
	6	—	—	—	—	— 0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[1, 5] = \begin{cases} \cancel{m[1, 1]}^0 + \cancel{m[2, 5]}^{2430} + 5 \times 10 \times 50 = 4930 \\ \cancel{m[1, 2]}^{150} + \cancel{m[3, 5]}^{930} + 5 \times 3 \times 50 = 1830 \\ \cancel{m[1, 3]}^{330} + \cancel{m[4, 5]}^{3000} + 5 \times 12 \times 50 = 6330 \\ \cancel{m[1, 4]}^{405} + \cancel{m[5, 5]}^0 + 5 \times 5 \times 50 = 1655 \end{cases}$$

After which, we can update the table:

m	j					
	1	2	3	4	5	6
i	1	0	150	330	630	1655
	2	—	0	360	330	2430
	3	—	—	0	180	930 1770
	4	—	—	—	0	3000 1860
	5	—	—	—	—	0 1500
	6	—	—	—	—	— 0

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[2, 6] = \begin{cases} \cancel{m[2, 2]} + \cancel{m[3, 6]} + 10 \times 3 \times 6 & = 1950 \\ \cancel{m[2, 3]} + \cancel{m[3, 6]} + 10 \times 12 \times 6 & = 2940 \\ \cancel{m[2, 4]} + \cancel{m[4, 6]} + 10 \times 5 \times 6 & = 2130 \\ \cancel{m[2, 5]} + \cancel{m[5, 6]} + 10 \times 50 \times 6 & = 5430 \end{cases}$$

After which, we can update the table:

m	j					
	1	2	3	4	5	6
i	1	0	150	330	630	1655
	2	—	0	360	330	2430
	3	—	—	0	180	930
	4	—	—	—	0	3000
	5	—	—	—	—	0
	6	—	—	—	—	—

Then we do the same for the following: Based on the lectures, we should approach it using matrix chain multiplication approach, which involves utilization of the following formula:

$$m[1, 6] = \begin{cases} \cancel{m[1, 1]} + \cancel{m[2, 6]} + 5 \times 10 \times 6 & = 2250 \\ \cancel{m[1, 2]} + \cancel{m[3, 6]} + 5 \times 3 \times 6 & = 2010 \\ \cancel{m[1, 3]} + \cancel{m[4, 6]} + 5 \times 12 \times 6 & = 2550 \\ \cancel{m[1, 4]} + \cancel{m[5, 6]} + 5 \times 5 \times 6 & = 2055 \\ \cancel{m[1, 5]} + \cancel{m[6, 6]} + 5 \times 50 \times 6 & = 3155 \end{cases}$$

After which, we can update the table:

m	j					
	1	2	3	4	5	6
i	1	0	150	330	630	1655
	2	—	0	360	330	2430
	3	—	—	0	180	930
	4	—	—	—	0	3000
	5	—	—	—	—	0
	6	—	—	—	—	—

“latex When doing a recursive lookup at the table and value k , we now can get the following:

- (a) for $m[1, 6] = m[1, 2] + m[3, 6] + \dots$, optimal $k = 2$.
- (b) for $m[3, 6] = m[3, 4] + m[5, 6] + \dots$, optimal $k = 4$.

Based on the following information, the optimal parenthesization is

$$((A_1A_2)((A_3A_4)(A_5A_6))) \Rightarrow (A_1A_2)(A_3A_4)(A_5A_6)$$

Problem 14-6 Professor Blutarsky is consulting for the president of a corporation that is planning a company party. The company has a hierarchical structure, that is, the supervisor relation forms a tree rooted at the president. The human resources department has ranked each employee with a conviviality rating, which is a real number. In order to make the party fun for all attendees, the president does not want both an employee and his or her immediate supervisor to attend.

Professor Blutarsky is given the tree that describes the structure of the corporation, using the left-child, right-sibling representation described in Section 10.3. Each node of the tree holds, in addition to the pointers, the name of an employee and that employee's conviviality ranking. Describe an algorithm to make up a guest list that maximizes the sum of the conviviality ratings of the guests. Analyze the running time of your algorithm.

Solution

The solution to this problem utilizes Dynamic Programming on a tree.

Recurrence Relation

For any given node, we have two choices to maximize the sum of the conviviality ratings:

- (a) If the node is included, we cannot include its immediate children. We must sum the node's conviviality with the optimal solutions for its grandchildren.
- (b) If the node is NOT included, we sum the optimal solutions for its immediate children.

This can be mathematically expressed as:

$$C[\text{node}] = \max \left(\begin{array}{l} \sum_{y \text{ is a child of node}} C[y] \\ \text{node.conviviality} + \sum_{y \text{ is a grandchild of node}} C[y] \end{array} \right)$$

Algorithm Description

To implement this efficiently using the provided **left-child, right-sibling** representation, we perform a recursive **post-order traversal** of the tree. This guarantees that before we calculate $C[\text{node}]$, the optimal values $C[y]$ for all its children and grandchildren have already been computed.

We use **memoization** by storing the computed maximum conviviality value in a new field, $\mathbb{C}$, on each node so it is only calculated once.

For each visited **node** in the post-order traversal:

- **Summing Children:** Follow the `node.left_child` pointer, then iterate through the `right_sibling` pointers until `null` is reached. Sum the stored `C` values of these nodes.
- **Summing Grandchildren:** While iterating through the children above, follow each child's `left_child` pointer and iterate through their respective `right_sibling` pointers. Sum the stored `C` values of these nodes.
- Apply the recurrence relation to find the maximum of the two choices.
- Store this maximum in `node.C`.

The final maximum conviviality rating for the entire company party will be stored in `root.C`.

Time Complexity

The time complexity of this procedure is $\Theta(n)$.

- Because we use memoization with a post-order traversal, the subproblem for each node is solved exactly once.
- At each node, finding the children and grandchildren involves traversing the `left_child` and `right_sibling` pointers.
- Across the entire algorithm, every pointer in the tree is followed a constant number of times (once when treating it as a child, and once when treating it as a grandchild).
- Since a tree with n nodes contains exactly $n - 1$ edges (pointers), the total amount of work done is directly proportional to the number of nodes. Therefore, the time complexity is $\Theta(n)$.